

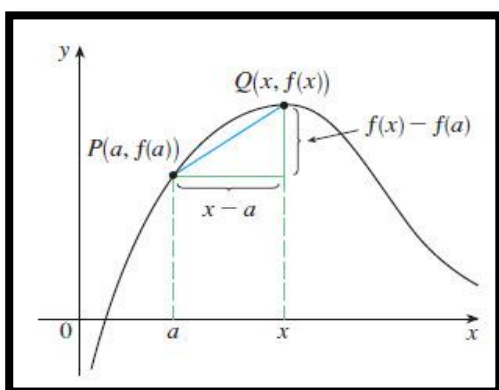
Mathematics

3- Summary for Differentiation

1. Introduction

If C is a curve with equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$, then we consider a nearby point $Q(x, f(x))$, where $x \neq a$ and compute the slope of the secant line PQ as shown in the figure,

$$m = \frac{f(x) - f(a)}{x - a} \quad \text{Slope of secant line}$$



As point Q approaches P , we will find the tangent at P . (As $Q \rightarrow P$ then $x \rightarrow a$) and that means

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad \text{Slope of tangent}$$

So, now m represents the slope of the tangent to the curve $y = f(x)$ at $x = a$.

We recall it as $f'(a)$ as

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

2. Definition of Differentiation

The first order derivative of $f(x)$ is denoted by $f'(x)$ or $\frac{df(x)}{dx}$ or $D(f(x))$, where $D = \frac{d}{dx}$ is called "Differential Operator".

2.1. Derivative of $f(x)$ at $x = a$

By putting $x - a = h$, we find that $x = h + a$ and when $x \rightarrow a$ then $h \rightarrow 0$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \equiv \text{Number}$$

2.2. General Definition

Instead of using a , we will use x so, we get

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \equiv \text{Function}$$

Note:

- In the previous definitions, you will get:
 - A **number** from the derivative at a point.
 - A **function** from the general definition

(If and only if the limit exists).

If the limit **does not exist (DNE)** or its result is $\pm\infty$, one says that the function is **not differentiable at this point** (using the definition of the derivative at a point) or it is **not differentiable at any x** (using the general definition).

- If $f(x)$ is differentiable at a point so, it is continuous at the same point.
- If $f(x)$ is not continuous at a point so, it is not differentiable at the same point.
- If x changes from x_0 to $x_0 + h$,

$$\text{Average rate of change} = \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0}$$

- Instantaneous rate of change @ $x = x_0$

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0)$$

3. Rules of Differentiation

$f(x)$	$f'(x)$
C <i>, C is a constant</i>	0
$u(x)^n$	$n u(x)^{n-1} * \dot{u}$
$C u(x)$ <i>, C is a constant</i>	$C \dot{u}(x)$
$u(x) \pm v(x)$	$\dot{u}(x) \pm \dot{v}(x)$
Product Rule: $u(x) * v(x)$	$\dot{u} * v + \dot{v} * u$
Quotient Rule : $\frac{u(x)}{v(x)}$	$\frac{\dot{u} * v - \dot{v} * u}{v^2}$
$\log_a u(x)$ <i>, a is constant</i>	$\frac{\dot{u}}{u(x)} * \frac{1}{\ln a}$
$a^{u(x)}$ <i>, a is constant</i>	$a^{u(x)} * \ln a * \dot{u}(x)$
$ u(x) $	$\frac{u(x)}{ u(x) } * \dot{u}(x)$

Chain Rule: if $y = g(u)$, $u = f(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{dx}$$

Trigonometric Functions

$f(x)$	$f'(x)$
$\sin(u(x))$	$\cos(u(x)) * \dot{u}$
$\cos(u(x))$	$-\sin(u(x)) * \dot{u}$
$\tan(u(x))$	$\sec(u(x))^2 * \dot{u}$
$\cot(u(x))$	$-\csc(u(x))^2 * \dot{u}$
$\sec(u(x))$	$\sec(u(x)) * \tan(u(x)) * \dot{u}$
$\csc(u(x))$	$-\csc(u(x)) * \cot(u(x)) * \dot{u}$

Inverse Trigonometric Functions

$f(x)$	$f'(x)$
$f(x) = \sin^{-1}(u(x))$	$\frac{\dot{u}}{\sqrt{1-u^2}}$ <i>, $u < 1$</i>
$f(x) = \cos^{-1}(u(x))$	$\frac{-\dot{u}}{\sqrt{1-u^2}}$ <i>, $u < 1$</i>
$f(x) = \tan^{-1}(u(x))$	$\frac{\dot{u}}{1+u^2}$
$f(x) = \cot^{-1}(u(x))$	$\frac{-\dot{u}}{1+u^2}$
$f(x) = \sec^{-1}(u(x))$	$\frac{\dot{u}}{ u \sqrt{u^2-1}}$ <i>, $u > 1$</i>
$f(x) = \csc^{-1}(u(x))$	$\frac{-\dot{u}}{ u \sqrt{u^2-1}}$ <i>, $u > 1$</i>

Hyperbolic Functions

$f(x)$	$f'(x)$
$\sinh(u(x))$	$\cosh(u(x)) * \dot{u}$
$\cosh(u(x))$	$\sinh(u(x)) * \dot{u}$
$\tanh(u(x))$	$\operatorname{sech}(u(x))^2 * \dot{u}$
$\coth(u(x))$	$-\operatorname{csch}(u(x))^2 * \dot{u}$
$\operatorname{sech}(u(x))$	$-\operatorname{sech}(u(x)) * \tanh(u(x)) * \dot{u}$
$\operatorname{csch}(u(x))$	$-\operatorname{csch}(u(x)) * \coth(u(x)) * \dot{u}$

Inverse Hyperbolic Functions

$f(x)$	$\dot{f}(x)$
$f(x) = \sinh^{-1}(u(x))$	$\frac{\dot{u}}{\sqrt{u^2 + 1}}$
$f(x) = \cosh^{-1}(u(x))$	$\frac{\dot{u}}{\sqrt{u^2 - 1}}$, $u > 1$
$f(x) = \tanh^{-1}(u(x))$	$\frac{\dot{u}}{1 - u^2}$, $ u < 1$
$f(x) = \coth^{-1}(u(x))$	$\frac{\dot{u}}{1 - u^2}$, $ u > 1$
$f(x) = \operatorname{sech}^{-1}(u(x))$	$\frac{-\dot{u}}{u\sqrt{1 - u^2}}$, $0 < u < 1$
$f(x) = \operatorname{csch}^{-1}(u(x))$	$\frac{-\dot{u}}{ u \sqrt{u^2 + 1}}$, $u \neq 0$

4. Applications of Differentiation

4.1. Logarithmic Differentiation

Case 1: If you have a relation $y = \frac{f(x) \cdot G(x) \cdot V(x)}{u(x)}$, you can find $\dot{y}$ by using the Quotient rule.

Or, for simplicity, take $(\ln \quad)$ for both sides you will get:

$$\ln y = \ln f(x) + \ln G(x) + \ln V(x) - \ln u(x)$$

Then, differentiate to get $\dot{y}$

Case 2: If $y = f(x)^{v(x)}$ and you need to find $\dot{y}$, take $(\ln \quad)$ for both sides you will get:

$$\ln y = V(x) \cdot \ln f(x)$$

Then, differentiate to get $\dot{y}$

4.2. Implicit Differentiation

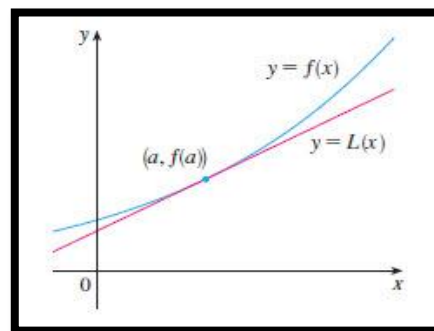
A **relation** is called implicit when you can't separate y from x .

If you are asked to find $\dot{y}$ for an implicit relation, follow the following procedure:

- Differentiate the given relation with respect to x .

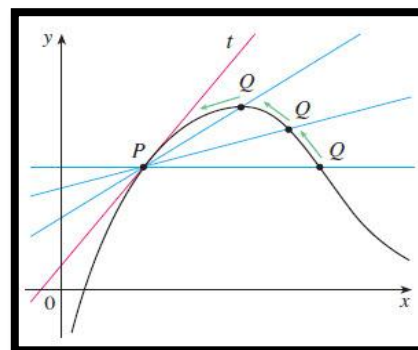
- Collect $\frac{dy}{dx}$ terms and take $\frac{dy}{dx}$ as a common factor from them.
- Solve for $\frac{dy}{dx}$.

4.3. Linearization



For a given curve $y = f(x)$, the tangent lines to this curve at $(a, f(a))$ will have a slope $\dot{f}(a)$.

But as we explained in the introduction of the differentiation, to find that tangent we had to assume a very nearby point with coordinates $(x, f(x))$, and this point is supposed to start moving towards $(a, f(a))$ until the secant line connecting the 2 points becomes the tangent line to this curve as shown in the following figure.



So, the line connecting $(a, f(a))$ and $(x, f(x))$ will have a slope = $\frac{f(x) - f(a)}{x - a}$. As we choose a very nearby point to the one which we need to find the tangent at, we can claim that the slope of the tangent line is approximately equal the slope of the secant line connecting $(a, f(a))$ and $(x, f(x))$ so, we get

$$\frac{f(x) - f(a)}{x - a} \approx \dot{f}(a) \text{ as } x \rightarrow a$$

$$\dot{f}(a) \cdot (x - a) \approx f(x) - f(a) \text{ as } x \rightarrow a$$

$$f(x) \approx f(a) + \dot{f}(a) \cdot (x - a) \text{ as } x \rightarrow a$$

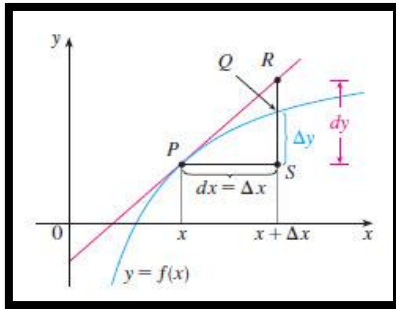
We define $L(x)$ as the standard linearization of the function $f(x)$ around the point $(a, f(a))$.

As for that we get,

$$L(x) = f(a) + f'(a) * (x - a) : @ x = a$$

And we say that $f(x) \approx L(x)$ as $x \rightarrow a$

4.4. The Differential



As shown in the above figure, the **blue** curve represents the function $y = f(x)$ and the **red** line represents the tangent to this curve at the point $P(x, f(x))$. For a small change $\Delta x = h$ in x , we would like to find the changes: " Δy & dy " in y on both the **blue** curve and the **red** line.

Note: $dx = \lim_{\Delta x \rightarrow 0}(\Delta x)$, i.e., for small values.

On the **blue** curve $y = f(x)$, the **change** Δy in y between the points **P** and **Q**, is defined as:

$$\Delta y = f(x + \Delta x) - f(x)$$

On the **red** line $L(x) = f(a) + f'(a) * (x - a)$, the change in y between the points **P** and **R**, is:

$$\Delta L = L(x + \Delta x) - L(x)$$

The **differential** is also denoted by **dy** where,

$$\begin{aligned} dy &= \lim_{\Delta x \rightarrow 0}(\Delta y) = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right) * \Delta x \\ &= \left(\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \right) * dx \\ &= f'(x) * dx \end{aligned}$$

And **dx** represents the change in x (i.e. $\Delta x \approx dx$, for small change).

4.5. L'Hopital Rule

It is commonly used to determine the value of the limits with "**Undetermined**" values.

Standard Case:

$$\text{If } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

If

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

, then

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)}$$

Special Case 1:

$$\text{If } \lim_{x \rightarrow a} f(x) * g(x) = 0 * \pm \infty$$

, rewrite $f(x) * g(x)$ as one as the following forms:

$$f(x) * g(x) = \frac{f(x)}{\frac{1}{g(x)}} \quad \text{or} \quad f(x) * g(x) = \frac{g(x)}{\frac{1}{f(x)}}$$

Then,

$$\lim_{x \rightarrow a} \frac{f(x)}{\frac{1}{g(x)}} \text{ or } \lim_{x \rightarrow a} \frac{g(x)}{\frac{1}{f(x)}} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

Special Case 2:

$$\text{If } \lim_{x \rightarrow a} [f(x) - g(x)] = \infty - \infty$$

, Try one of the following:

- If $f(x)$ & $g(x)$ have different denominators, **make a common denominator** and then continue as the standard case.
- If $f(x)$ & $g(x)$ have no denominators, take $f(x)$ or $g(x)$ as a common factor "function" then,

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) * \left(1 - \frac{g(x)}{f(x)} \right)$$

Then for

$$\lim_{x \rightarrow a} \frac{g(x)}{f(x)}$$

, apply the standard case.

Special Case 3:

If $\lim_{x \rightarrow a} f(x)^{g(x)} = 0^0 \text{ or } \infty^0 \text{ or } 1^\infty$

, Proceed as the following:

- Let $y = f(x)^{g(x)}$
- Take $(\ln)$ for both sides and you will get,
 $\ln y = g(x) * \ln f(x)$
- Calculate

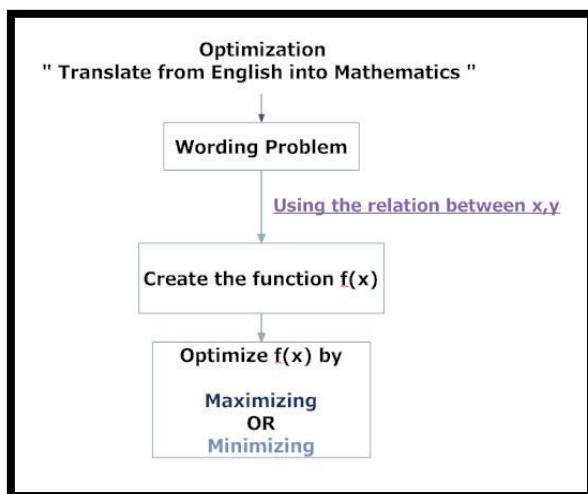
$$\lim_{x \rightarrow a} g(x) * \ln f(x) = m$$

- $\lim_{x \rightarrow a} f(x)^{g(x)} = e^m$

4.6. Extrema and Inflections

[See next page](#)

4.7. Optimization



4.8. Curve Sketching

- Determine the domain of the function.
- Find y-intercept (@ $x=0$) and x-intercept (@ $y=0$).
- Study the symmetry of the function (Even-Odd-Neither).
- Find the asymptote lines of the function.
- Using $f'(x)$,
 - Determine the intervals of increasing and decreasing.
 - Find Local maximums and Local minimums.
- Using $f''(x)$, Study Concavity and determine the inflection points.

What does it means "Symmetry"?

- A function $f(x)$ is called **ODD** iff $f(-x) = -f(x)$
That means $f(x)$ is symmetric about that origin. "i.e. the graph of $f(x)$ is in the 1st and 3rd quad OR it is in 2nd and 4th quad".

Like:

$$f(x) =$$

$$x, x^3, x^5, \dots, \frac{1}{x}, \sin x, \sinh x, \tan x, \tanh x, \cot x, \coth x$$

- A function $f(x)$ is called **EVEN** iff $f(-x) = f(x)$
That means $f(x)$ is symmetric about that y-axis (mirror of the function). "i.e. the graph of $f(x)$ is in the 1st and 2nd quad OR it is in 3rd and 4th quad".

$$\text{Like: } f(x) = x^2, x^4, \dots, \frac{1}{x^2}, \cos x, \cosh x, \sec x, \operatorname{sech} x$$

Remark: for the product of two functions

- Even function * Even function = Even
- Even function * Odd function = Odd
- Odd function * Odd function = Even

What does it means "Asymptote Line"?

- ❖ A **horizontal** asymptote " $y = L$ " of $f(x)$ is obtained by:

$$\lim_{x \rightarrow \infty^\pm} f(x) = L$$

- ❖ If $f(x)$ is a **fractional** function: a **vertical** asymptote " $x = a$ " of $f(x)$ is obtained by:

Equating the "**Deno** of $f(x)$ to 0 and we must have

$$\lim_{x \rightarrow a^\pm} f(x) = \pm \infty$$

- ❖ If $f(x)$ is a **fractional** function:

$$\text{Degree of Num} = \text{Degree of Deno} + 1$$

One can find the **oblique** asymptote line by **Long Division**.

In this case:

- " $y = \text{quotient}$ " is the oblique asy. line equation.
- $f(x)$ will be written as:

$$f(x) = \text{Oblique} + \frac{\text{remainder}}{\text{Deno}}$$

Given $f(x)$

Extrema points of $f(x)$

1] Get the Critical points of $f(x)$

Find x : Such that $f'(x)=0$ or $f'(x)$ is undefined

$$f'(x) = \frac{\square \rightarrow 0}{\Delta \rightarrow 0} \Rightarrow \text{get } x=c \text{ "critical pts."}$$

2] Check if $c \in D_f$ (or in the given domain)

Inflection points of $f(x)$

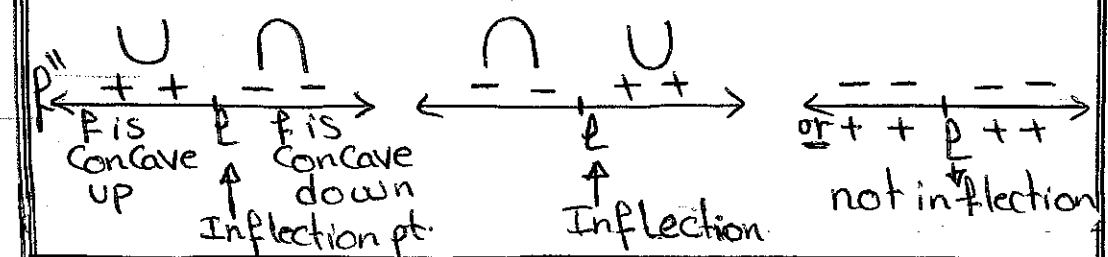
1] Get the Critical points of $f'(x)$

Find x : $f''(x)=0$ or $f''(x)$ is undefined

$\Rightarrow$ get $x=c$

2] Check if $x=c \in D_{f'}$

Then Line Test for f''



Absolute Extrema

$f(x)$ must be defined on a closed interval $[a,b]$

Then Find The values of $f(c)$, $f(a)$ & $f(b)$

Abs. Max $\Rightarrow$ is the highest value

Abs. Min $\Rightarrow$ is the smallest value.

Special Case.

If $f(x)$ is defined on an open interval $]a,b[$ then you can get

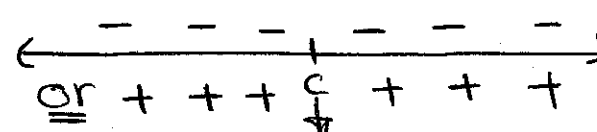
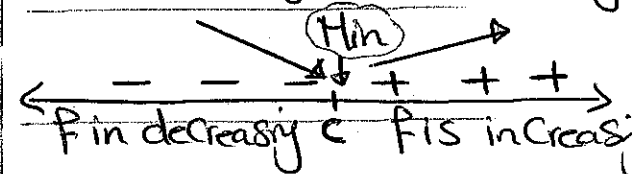
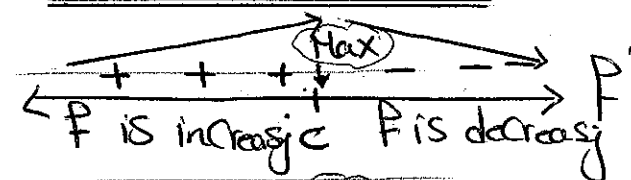
Abs. extrema if $f(x)$ has only one critical point $x=c$, to decide if it's Abs. Max or

Abs. Min, you have to study $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$

Relative Extrema

OR

1st derivative Test



Neither Max nor Min

2nd derivative Test

$f''(c) < 0 \Rightarrow x=c$ is relative Max.

$f''(c) > 0 \Rightarrow$ " " " Min.

$f''(c) = 0 \Rightarrow$ Test fails

$\Rightarrow$ No Conclusion

$\Rightarrow$ you have to use 1st deriv. Test to decide.

Some important points of a function $f(x)$

Extrema Points

Given $y=f(x)$

Find $f'(x)$

$f'(x)$
fractional ?

$f'(x) = 0$ ← No

Yes → $\text{Num}(f')=0$
 $\text{Deno}(f')=0$

The points "C" you get are Critical Points of f ... Check if $C \in \text{Domain}(f)$

Judge for C

Absolute Extrema

If a closed interval $[a,b]$ is given

1- Find $f(a), f(b), f(c)$

*The highest value of them is (Absolute max.)
*The lowest value of them is (Absolute min.)

Local Extrema

1st Derivative test

Test sign of f'

* f' changes from (+) to (-) : Local max.
* f' changes from (-) to (+) : Local min.
{ if sign of f' doesn't change : Neither }
* if f' is (+ve) $\Rightarrow f$ is INCREASING
* if f' is (-ve) $\Rightarrow f$ is DECREASING

2nd Derivative test

Test $f''(c)$

* $f''(c) < 0$... Local max.
* $f''(c) > 0$... Local min.
* $f''(c) = 0$... Test fails
{ If the test fails, Use the "1st Derivative test" }

Inflection Points

Given $y=f(x)$

Find $f'(x)$

Find $f''(x)$

$f''(x)$
fractional ?

$f''(x) = 0$ ← No

Yes → $\text{Num}(f'')=0$
 $\text{Deno}(f'')=0$

The points "d" you get are Critical Points of f' ... Check if $d \in \text{Domain}(f)$ and $\text{Domain}(f')$

Judge for d

Test of Concavity

Test sign of f''

* sign of (f'') changes from (+) to (-) : Inflection Point
* sign of (f'') changes from (-) to (+) : Inflection Point
{ if sign of f'' doesn't change : Not an inflection point }

* if f'' is (+ve) $\Rightarrow f$ is Concave UP
* if f'' is (-ve) $\Rightarrow f$ is Concave DOWN

Inflection Points "d" of the given function $f(x)$

are the accepted critical points "d" of $f'(x)$

IFF

* sign of (f'') changes from
(+) to (-)
OR
(-) to (+)

GUIDELINE FOR SOLVING OPTIMIZATION PROBLEMS

1. **Read each problem slowly and carefully.** Read the problem at least two times before trying to solve it.
2. If appropriate, **draw a sketch or diagram of the problem** to be solved.
3. **Define variables** to be used and carefully label your diagram with these variables. This step is very important because it leads directly or indirectly to the creation of mathematical equations.
4. **Write down all equations which are related to your problem or diagram.** Clearly denote the “target” function which you are asked to maximize or minimize. Experience will show you that **MOST of optimization problems will begin with two equations.** One equation is **a "constraint" equation** and the other is the **"optimization" function.** The "constraint" equation is used to solve for one of the variables. This is then substituted into the "optimization" function before differentiation occurs. Some problems may have NO constraint equation. Some problems may have two or more constraint equations.
5. **Before differentiating, make sure that the optimization equation is a function of one variable ONLY.** Then differentiate using the well-known rules of differentiation.
6. Verify that your result is **a maximum or minimum** value using the first or second derivative test for extrema.